GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Network Theory — Essential GATE Formula Sheet & Circuit Theorems

KCL/KVL, Two-Port Conversions, Transient Step Response, RLC Resonance & AC Power

GATE Section: Section 2 • **Marks:** 8 - 11 Marks • **Core Laws:** Ohm, Thevenin, Norton, Phasors

CRITICAL CIRCUIT FORMULAS & SHORTCUTS
- **First-Order Transients:** v(t) = v(inf) + [v(0+) - v(inf)] * exp(-t / tau), where tau = R_th * C or L / R_th.
- **Series RLC Resonance:** omega_0 = 1/sqrt(L*C), Q = (1/R)*sqrt(L/C) = omega_0*L/R, Bandwidth = R/L rad/s.
- **Parallel RLC Resonance:** omega_0 = 1/sqrt(L*C), Q = R*sqrt(C/L) = R/(omega_0*L), Bandwidth = 1/(R*C) rad/s.

SECTION: FUNDAMENTAL THEOREMS & DC/AC SOLUTIONS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Ohm's & Kirchhoff's	**sum(I_in) = sum(I_out) (KCL)** **sum(V_drops) = 0 (KVL)**	Conservation of charge (KCL) Conservation of energy (KVL)	KCL and KVL are valid for linear, non-linear, time-variant, and time-invariant networks.
Thevenin's Theorem	**V_th = open-circuit voltage Voc** **R_th = Voc / Isc = V_test / I_test**	R_th with dependent sources: Deactivate independent sources, apply 1V or 1A test source	R_th can be negative in circuits containing active dependent sources!
Maximum Power (DC)	**P_max = V_th^2 / (4 * R_th)**	Condition: R_L = R_th Efficiency = 50% at max power	If R_L is variable and R_th is fixed: R_L = R_th. If R_th is variable and R_L is fixed: R_th = 0.
Maximum Power (AC)	**Z_L = Z_th* = R_th - j*X_th** **P_max = |V_th|^2 / (4 * R_th)**	Complex conjugate match If Z_L purely resistive: R_L = |Z_th|	If load angle is fixed: |Z_L| = |Z_th|.
Star-Delta Transform	**R_delta = R_a + R_b + (R_a * R_b / R_c)** **R_star = (R_1 * R_2) / (R_1 + R_2 + R_3)**	Identical resistors: R_delta = 3 * R_star	Saves time in symmetrical bridge networks.

SECTION: TRANSIENTS & SECOND-ORDER RLC DYNAMICS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Inductor & Capacitor Laws	**v_L(t) = L * di/dt** **i_C(t) = C * dv/dt**	Inductor current cannot change instantaneously: i_L(0+) = i_L(0-) Capacitor voltage cannot change: v_C(0+) = v_C(0-)	At t = 0+: Uncharged C is a SHORT circuit, unenergized L is an OPEN circuit. At t = inf (DC): C is OPEN, L is SHORT.
Universal 1st-Order Step	**x(t) = x(inf) + [x(0+) - x(inf)] * e^(-t / tau)**	tau = R_th * C (RC circuit) tau = L / R_th (RL circuit)	Find x(0+), find x(inf), find Thevenin resistance R_th seen by storage element.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Series RLC Damping	**s^2 + (R/L)*s + (1/LC) = 0** **zeta = R / (2 * sqrt(L/C))** **omega_n = 1 / sqrt(LC)**	zeta > 1: Overdamped zeta = 1: Critically damped (fastest without overshoot) zeta < 1: Underdamped (oscillatory)	Critical resistance: R_crit = 2 * sqrt(L / C).
Parallel RLC Damping	**s^2 + (1/(RC))*s + (1/LC) = 0** **zeta = 1 / (2 * R * sqrt(C/L))**	zeta > 1: Overdamped zeta = 1: Critically damped	Critical resistance: R_crit = 0.5 * sqrt(L / C).

SECTION: TWO-PORT PARAMETERS & CONVERSION MATRIX

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Z-Parameters (Impedance)	**V1 = Z11*I1 + Z12*I2** **V2 = Z21*I1 + Z22*I2**	Z11 = V1/I1 | I2=0, Z12 = V1/I2 | I1=0 Reciprocal: Z12 = Z21, Symmetric: Z11 = Z22	Series-Series interconnected two-port: Z_total = Z_A + Z_B.
Y-Parameters (Admittance)	**I1 = Y11*V1 + Y12*V2** **I2 = Y21*V1 + Y22*V2**	Y11 = I1/V1 | V2=0, Y12 = I1/V2 | V1=0 Reciprocal: Y12 = Y21, Symmetric: Y11 = Y22	Parallel-Parallel interconnected two-port: Y_total = Y_A + Y_B.
h-Parameters (Hybrid)	**V1 = h11*I1 + h12*V2** **I2 = h21*I1 + h22*V2**	h11 = input imp, h12 = reverse voltage gain h21 = forward current gain, h22 = output adm	Reciprocal: h12 = -h21, Symmetric: det(h) = h11*h22 - h12*h21 = 1.
ABCD Parameters (Transmission)	**V1 = A*V2 - B*I2** **I1 = C*V2 - D*I2**	A = V1/V2 | I2=0, B = -V1/I2 | V2=0 C = I1/V2 | I2=0, D = -I1/I2 | V2=0	Reciprocal: A*D - B*C = 1, Symmetric: A = D. Cascade connection: T_total = T_A * T_B.